Thermodynamic description of the Mg–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Mg–Sn binary system as calculated with pycalphad from the SGTE dataset MgSn-04Fri.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Mg) + Mg2Sn, lies at 563.4 °C.

1. Introduction

The Mg–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as congruent compound. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of S.G. Fries, unpublished work, 2004. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Mg melts at 649.9 °C. The melting point of Sn reproduced by the dataset is 231.9 °C.

2. Phase diagram

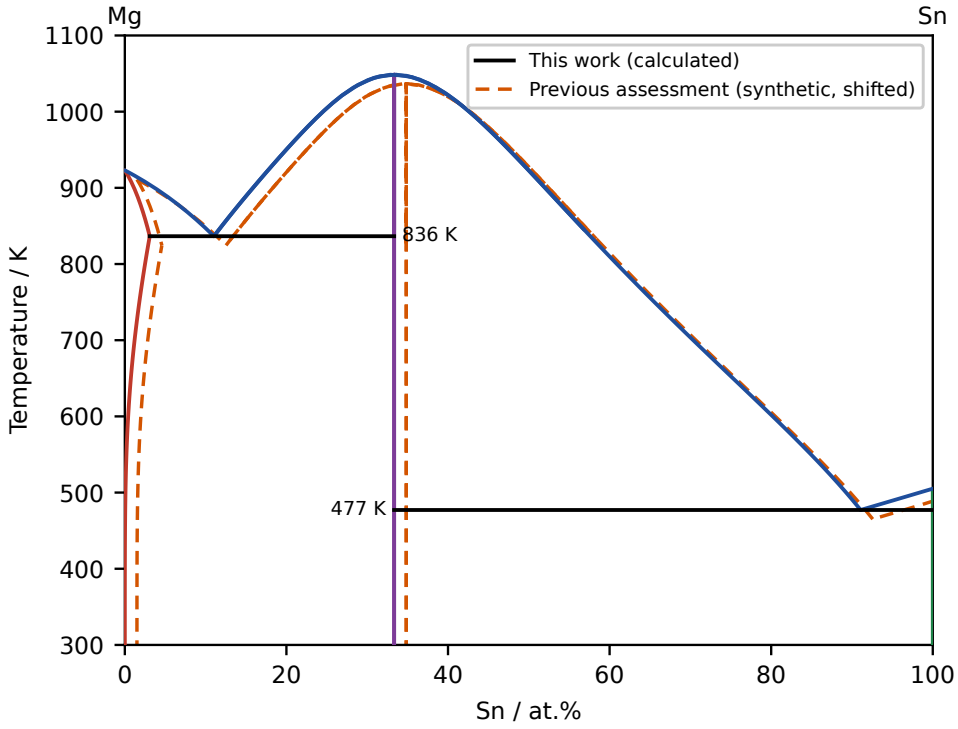


Fig. 1. Calculated Mg–Sn phase diagram and a dashed, deliberately shifted second assessment.

The dataset places a eutectic (L → (Mg) + Mg2Sn) at 563.4 °C; the composition of L is x(Sn) = 0.110. We obtain 204.0 °C for the eutectic reaction (L → Mg2Sn + (Sn)), where L has a mole fraction of 0.911. Mg2Sn melts congruently at 775.5 °C and x(Sn) = 0.333.

Table 1. Invariant reactions in the Mg–Sn system (calculated).

Reaction	Type	T / K	Compositions (at.% Sn)		
			phase 1	phase 2	phase 3
L → (Mg) + Mg2Sn	eutectic	836.5	11.0	3.1	33.3
L → Mg2Sn + (Sn)	eutectic	477.1	91.1	33.3	100.0

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1150 K the integral mixing enthalpy of the liquid has its minimum, −14298 J/mol, at x(Sn) = 0.380. The calculated enthalpy of formation of Mg2Sn at 298.15 K is −25.64 kJ/mol per mole of atoms.

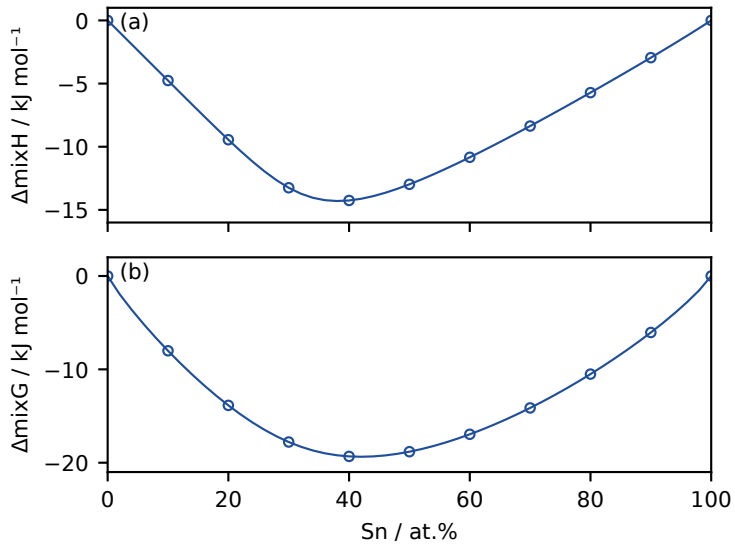


Fig. 2. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1150 K.

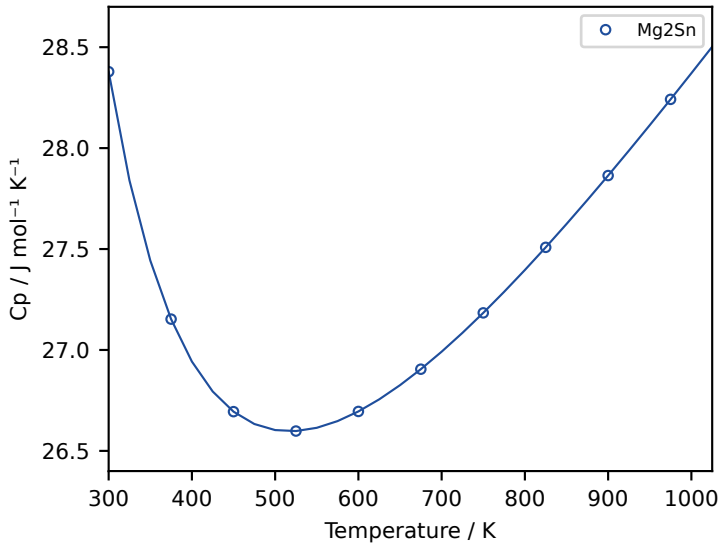


Fig. 3. Calculated heat capacity of Mg2Sn.

Table 2. Calculated formation properties of the intermediate phases (Mg–Sn).

		Calculated at 298.15 K, per mole of atoms	
Phase	**at.% Sn**	**ΔfH / kJ mol⁻¹**	**S / J mol⁻¹ K⁻¹**
Mg2Sn	33.3	−25.64 ± 0.5	35.17

Reference: stable elements (SER) at 298.15 K. ± values are synthetic illustration uncertainties, not data.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.